Math140C - HW #6

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Q1

Let

$$A := \{x \in E \mid f(x) > 0\}$$
$$E_n := \{x \in E \mid f(x) > 1/n\}, \quad (n \in \mathbb{Z}_+).$$

Then

$$A = \bigcup_{n=1}^{\infty} E_n.$$

Since the measure μ is monotone and countably additive,

$$\mu(A) = 0 \iff \forall n \in \mathbb{Z}_+,\ \mu(E_n) = 0. \tag{1}$$

If $f(x)$ is not zero almost everywhere on E, then $\mu(A) > 0$. By (1), there is some $n \in \mathbb{Z}_+$ such that $\mu(E_n) > 0$. By definition of E_n,

$$\frac{1}{n}\chi_n < f$$

on E. Hence

$$\begin{aligned} \frac{1}{n}\mu(E_n) &= \int_E \frac{1}{n}\chi_n d\mu \\ &\leq \int_E f d\mu, \end{aligned}$$

which means $\int_E f d\mu > 0$. This contradicts the assumption that $\int_E f d\mu = 0$, so it must be that $\mu(A) = 0$; that is, $f(x) = 0$ almost everywhere on E.

Q2

E is a measurable subset of E, so if $f \geq 0$, then $f(x) = 0$ almost everywhere on E by Q1. Suppose there is some $x \in E$ such that $f(x) < 0$. In other words, letting

$$A^{nn} = \{x \in E \mid f(x) \leq 0\},$$

$A^{np} \neq \emptyset$. f is a measurable function, so A^{np} is a measurable set. By assumption on the measurable subsets of E,

$$0 = \int_{A^{np}} f d\mu = \int_{A^{np}} f^+ d\mu - \int_{A^{np}} f^- d\mu \tag{2}$$

$f^+ = 0$ on A^{np}, so (2) gives $\int_{A^{np}} f^- d\mu = 0$. Of course, f^- is nonnegative on A^{np}, so by Q1,

$$f^-(x) = 0 \text{ almost everywhere on } A^{np}. \tag{3}$$

Let $A^- = \{x \in E \mid f(x) < 0\}$. Clearly, $A^- \subset A^{np}$. Hence the conclusion (3) implies $\mu(A^-) = 0$. Symmetric argument gives $\mu(A^+) = 0$ where $A^+ := \{x \in E \mid f(x) > 0\}$. It follows that $f(x) = 0$ almost everywhere on E.

Q3

Let

$$h(x) = \liminf_{n\to\infty} f_n(x)$$
$$g(x) = \limsup_{n\to\infty} f_n(x)$$

Since each f_n is a measurable function, h and g are measurable functions. Recall that a sequence has a limit if and only if its liminf and limsup are equal. Hence for a given point $x \in X$,

$$(f_n(x))_{n\in\mathbb{Z}_+} \text{ converges} \iff h(x) = g(x) \iff h(x) - g(x) = 0 \tag{4}$$

Let $k = h - g$. k is a measurable function. By (4),

$$\begin{aligned}\{x \in X \mid (f_n(x))_{n\in\mathbb{Z}_+} \text{ converges}\} &= \{x \in X \mid h(x) = 0\} \\ &= \{x \in X \mid h(x) \geq 0\} \cap \{x \in X \mid h(x) \leq 0\}\end{aligned} \tag{5}$$

and (5) is measurable; the set of points such that $(f_n(x))$ converges is measurable.

Q6

It is clear that $f_n \to 0$ uniformly: fix an $\epsilon > 0$ and pick $N \in \mathbb{Z}_+$ such that $1/N < \epsilon$. Then for all $n \geq N$ and for all $x \in \mathbb{R}$,

$$|f_n(x)| \leq 1/N < \epsilon.$$

However, for every $n \in \mathbb{Z}_+$,

$$\begin{aligned}\int_{-\infty}^{\infty} f_n dx &= \int_{-n}^{n} f_n dx \\ &= \int_{-n}^{n} \frac{1}{n} dx = 2,\end{aligned}$$

so Lebesgue's dominated convergence theorem is not guaranteed for uniformly convergent sequences of functions even though uniformly convergent sequences are uniformly bounded, i.e., dominated by a single constant function.

Q8

$$f \in \mathscr{R}_a^b \iff f \text{ is continuous almost everywhere on } [a, b],$$

and for points $x \in [a, b]$ where f is continuous,

$$F'(x) = f(x).$$

So assuming $f \in \mathscr{R}_a^b$, the containment

$$\{x \in [a, b] \mid F'(x) = f(x)\} \supset \{x \in [a, b] \mid f \text{ is continuous on } x\}$$

shows $F'(x) = f(x)$ almost everywhere on $[a, b]$.